The estimated Van't Hoff factor for acetic acid solution in Benzene is 0.5 and that of sodium chloride solution in water is 2. The degree of dissociation or association can be related to Van't Hoff factor (i) using the following relationships

$$\alpha_{dissociation} = \frac{i-1}{n-1}$$

(Where n is number ions /species formed by the dissociation of a single molecule)

$$\alpha_{association} = \frac{(1-i)n}{n-1}$$

(here, n is the number of solute involved in association.

The equations relating the four colligative properties with the concentration of the solutes can be rewritten as follows by incorporating the van't Hoff factor

Relative lowering of vapour pressure,

$$= \frac{p^o_{solvent} - p_{solution}}{p^o_{solvent}} = i \frac{n_{solute}}{n_{solvent}}$$

Elevation of boiling point,

$$\Delta T_b = iK_b m$$

Depression in the freezing point,

$$\Delta T_f = iK_f m$$

Osmotic pressure,

$$\pi = i \frac{w_{solute}}{V} \frac{RT}{M_{solute}}$$

For a solute that does not dissociate or associate the van't Hoff factor is equal to 1 (i = 1) and the molar mass will be close to the actual molar mass.

For the solutes that associate to form higher oligomers in solution the van't Hoff factor will be less than one (i <1) and the observed molar mass will be greater than the actual molar mass.

For solutes that dissociates into their constituent ions the van't Hoff factor will be more than one (i > 1) and the observed molar mass will be less than the normal molar mass.

Example Problem - 7

The depression in freezing point is 0.24K obtained by dissolving 1g NaCl in 200g water. Calculate van't-Hoff factor. The molal depression constant is 1.86 K Kg mol^{-1}

Molar mass of solute

$$= \frac{1000 \times K_f \times \text{mass of NaCl}}{\Delta T_f \times \text{mass of solvent}}$$

$$= \frac{1000 \times 1.86 \times 1}{0.24 \times 200}$$

$$= 38.75 g\ mol^{-1}$$

$$= 38.75\ g\ mol^{-}$$

Theoretical molar mass of NaCl is =

$$i = \frac{\text{Theoretical molar mass}}{\text{Experimental molar mass}} = \frac{58.5}{38.75}$$

$$= 1.50$$

Evaluate Yourself

14. 0.2 m aqueous solution of KCl freezes at -0.68°C calculate van't Hoff factor. k_f for water is 1.86 K kg mol^{-1}.

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